

OpenDccCLib — Architecture (As-Built)

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*Verified against the source at 2026-09-25 (commit 65d4570, followed by a full documentation audit against `src/dcc`). This document describes the library **as it is implemented today**. For the original design intent (which predates the role-first naming refactor), see `archive/OpenDccCLib_Requirements.md`. For what is done vs. pending, see `ComplianceOverview.md`.*

1. Overview

OpenDccCLib is a portable C library implementing NMRA DCC (S-9.1 through S-9.3.2) for both command-station and decoder roles. Design constraints:

- **No dynamic memory.** All buffers and state are statically sized by `USER_DEFINED_DCC_*` constants.
- **No OS / RTOS.** Runs from a main loop plus a small number of ISRs.
- **Dependency injection everywhere.** Cross-module calls go through `interface_dcc_*_t` function-pointer structs, never direct includes; the utilities modules are the documented exemption (`dcc_railcom_utilities` is called directly by the two RailCom modules). `dcc_config.c` (the wiring module) includes the headers of every module it wires.
- **Portable hardware layer.** The library never touches hardware directly; the user supplies driver callbacks in one `dcc_config_t` struct.

The user includes `dcc_config.h`, populates one struct, and calls `DccConfig_initialize()`; the application API comes from the `dcc_application_*.h` headers.

2. Feature flags

Defined by the user in `dcc_user_config.h`, validated with `#error/#warning` in `dcc_config.h` and `dcc_types.h`.

Flag	Enables
<code>DCC_COMPILE_COMMAND_STATION</code>	Packet encoding, bit framing, scheduler, service-mode core; with <code>DCC_COMPILE_RAILCOM</code> the RailCom cutout + receive
<code>DCC_COMPILE_DECODER</code>	Packet decoding, CV storage, fail-safe; with <code>DCC_COMPILE_RAILCOM</code> the RailCom transmit engine
<code>DCC_COMPILE_ACCESSORY_DECODER</code>	Reserved for the accessory-decoder RailCom feature (not implemented); today it only enables the RailCom 4/8 encoders and satisfies the role check. Packet reception needs <code>DCC_COMPILE_DECODER</code>
<code>DCC_COMPILE_RAILCOM</code>	Every RailCom module, config field and function; must be paired with a role flag (<code>dcc_types.h</code>)
<code>DCC_COMPILE_SERVICE_MODE_DIRECT</code>	Direct byte/bit programming (requires CS)
<code>DCC_COMPILE_SERVICE_MODE_PAGED</code>	Paged programming (requires CS)
<code>DCC_COMPILE_SERVICE_MODE_REGISTER</code>	Physical register programming (requires CS)
<code>DCC_COMPILE_SERVICE_MODE_ADDRESS</code>	Address-only programming (requires CS)
<code>DCC_COMPILE_SERVICE_MODE_TASK_{DIRECT, PAGED, REGISTER, ADDRESS, DETECT}</code>	The read/write/verify orchestrators and mode detection on top of the primitives; not cross-checked against their primitive

RailCom and each service-mode method are compile-time choices; hardware is optional at runtime. `DCC_COMPILE_RAILCOM` strips every RailCom module, field and function. The four primitive flags each require `DCC_COMPILE_COMMAND_STATION` (`#error`); the `TASK_*` flags are not cross-checked. Within a compiled feature, missing hardware is tolerated: RailCom stays inert until `main_track.railcom` and `railcom_timer_start` are wired, and decoder transmit until `railcom_tx_pin_set` is. Service mode without `current_sense_read` still transmits and finishes `NO_ACK` or `ERROR`. A device may define both CS and `DECODER` for booster/repeater use; `dcc_config.h` warns when no role is enabled.

3. User-configurable constants

Set in `dcc_user_config.h`:

- `USER_DEFINED_DCC_SCHEDULER_SLOT_COUNT` — max concurrent active scheduler slots
- `USER_DEFINED_DCC_MAX_LOCOS` — locomotives tracked by the application's loco table
- `USER_DEFINED_DCC_PREAMBLE_BITS_OPS` — operations-mode preamble (≥ 14 ; ≥ 16 with RailCom)
- `USER_DEFINED_DCC_RAILCOM_BUFFER_DEPTH` — RailCom receive ring-buffer depth
- `USER_DEFINED_DCC_SERVICE_MODE_RETRIES`
- `USER_DEFINED_DCC_ACK_THRESHOLD_MA`, `..._ACK_MIN_DURATION_US`, `..._ACK_MAX_DURATION_US`; `..._ACK_DROP_OUT_TOLERANCE_US` is optional (default 116)
- `USER_DEFINED_DCC_DECODER_PACKET_QUEUE_DEPTH` (≥ 2 , one slot reserved; a full queue drops the newest packet). F0–F68 are always dispatched; there is no function-count constant

Every count is validated ≥ 1 ; the command-station constants are required only with `DCC_COMPILE_COMMAND_STATION`; a preamble of 16–17 with RailCom raises a `#warning`. The refresh pacing constants `DCC_REFRESH_PROMPT_SENDS`, `DCC_REFRESH_COLD_CYCLES`, `DCC_REFRESH_COLD_MAX_CYCLES` and `DCC_REFRESH_CV11_FLOOR` are `#ifndef` defaults in `dcc_defines.h` that `dcc_user_config.h` may override, with `#error` guards on their relationships.

One-shot packet repeat counts are not user constants: the builders set them from the `DCC_REPEAT_*` table in `dcc_defines.h` (CV write 2, verify 1, date 3, time 1, accessory NOP/stop 1, everything else 2); an application may overwrite `repeat_count` after a builder returns, and 0 means the scheduler never sends the packet.

4. The `dcc_config_t` struct

One user-facing struct (`dcc_config.h`). Required fields must be non-NULL; NULL-optional fields disable the associated feature at runtime.

Common (always required): `lock_shared_resources`, `unlock_shared_resources`, `get_timestamp_usec`.

Command Station: a shared fixed-period DCC timer (`shared_timer_start/stop`, clocked at `DCC_ONE_BIT_HALF_PERIOD_US = 58 µs`). Per-channel hardware is described by `dcc_output_hw_t` for `main_track` and `service_track`: `pin_toggle`, `track_power_set` (called by `power_on/off` on both tracks and by `enter/exit_service_mode` on the service track), optional `current_sense_read` (read on the service track only, for ACK sampling) and an optional nested `dcc_railcom_hw_t` (read on the main track only). There is no per-channel timer; both channels run from the shared timer. With `DCC_COMPILE_RAILCOM`: a RailCom one-shot timer (`railcom_timer_start/stop`), the five cutout periods (`railcom_cutout_start_delay_us`, `railcom_uart_rx_delay_us`, `railcom_ch1_window_us`, `railcom_ch1_ch2_gap_us`, `railcom_ch2_window_us`; 0 = spec default), and `dcc_railcom_hw_t` (`begin_railcom_cutout`, `end_railcom_cutout`, `uart_rx_enable`, `uart_rx_disable`, `uart_read`, `on_railcom_datagram_result`). Optional callback: `on_packet_sent`. Service-mode results are delivered per call through the task callbacks (`on_complete`, `on_progress`, `on_detect`), not through a config callback.

Decoder: required `cv_read/cv_write/cv29_apply_supported_features` (the last tolerates NULL); NULL-optional `factory_reset`, `cv_read_indexed/cv_write_indexed`, `start_ack_pulse/stop_ack_pulse`, and with `DCC_COMPILE_RAILCOM` `railcom_tx_pin_set + railcom_delay_us` (required once the pin is set) + `on_railcom_request`, which runs on the edge path rather than the main loop. Twelve `on*_command` notification callbacks (`speed`, `emergency_stop`, `function`, `accessory_basic`, `accessory_extended`, `cv_write`, `cv_verify`, `cv_bit`, `consist`, `binary_state_short`, `binary_state_long`, `analog_function`) plus `on_failsafe_entered/exited`. The DCC edge interrupt is masked during a RailCom transmit through `lock_shared_resources`; there is no separate edge-IRQ hook. Note the CV callbacks (`on_cv_write_command`, `on_cv_verify_command`, `on_cv_bit_command`) carry a `bool service_mode` argument so the application can distinguish service-mode programming from POM.

5. Public API surface

Lifecycle & ISR entry points (`dcc_config.h`)

- `DccConfig_initialize(const dcc_config_t *)`
- `DccConfig_run()` — main loop; **all application callbacks fire from here, except the decoder's `on_railcom_request`, which runs on the edge path**
- `DccConfig_58us_timer_isr()` — [CS] shared bit timer
- `DccConfig_railcom_one_shot_timer_isr()` — [CS + RAILCOM] RailCom cutout state machine
- `DccConfig_set_railcom_cutout_timing()`, `DccConfig_cancel_railcom_cutout()`, `DccConfig_railcom_cutout_is_active()` — [CS + RAILCOM] runtime cutout control
- `DccConfig_100ms_timer_tick()` — [CS] reserved housekeeping hook; does nothing in this release
- `DccConfig_decoder_edge_isr(uint32_t timestamp_usec)` — [DECODER] input-capture edge
- `DccConfig_reload_address_cvs()` — [DECODER] re-read the address CVs after the application writes its storage outside the library

Application modules (role-first naming)

Module	Role	Public prefix
<code>dcc_application_command_station_main_track</code>	CS	<code>DccApplicationCommandStationMainTrack_</code> — <code>power_on/off</code> , <code>send_packet</code> , <code>add_to_auto_refresh</code> , <code>remove_from_auto_refresh</code> , <code>remove_all_auto_refresh</code>
<code>dcc_application_command_station_service_track</code>	CS	<code>DccApplicationCommandStationServiceTrack_</code> — <code>power_on/off</code> , <code>enter/exit_service_mode</code> , <code>is_service_mode_active</code> ; <code>direct_{read,write}_{cv,bit}</code> , <code>paged_{read,write}_{cv,bit}</code> , <code>register_{read,write}_{cv,bit}</code> , <code>register_verify_value</code> , <code>register_factory_reset</code> , <code>address_{read,write,verify}</code> , <code>address_{read,write}_bit</code> , <code>detect_mode</code> , each group gated by its <code>DCC_COMPILE_SERVICE_MODE_TASK_*</code> flag
<code>dcc_application_command_station_packet</code>	CS	<code>DccApplicationCommandStationPacket_load_*</code> — packet builders
<code>dcc_application_decoder_cv</code>	DECODER	<code>DccApplicationDecoderCv_</code> — <code>read</code> , <code>write</code> , <code>is_locked</code>
<code>dcc_application_decoder_railcom</code>	RAILCOM + DECODER	<code>DccApplicationDecoderRailcom_</code> — <code>pom_response</code> , <code>dynamic_data</code> , <code>cv_auto_transfer</code> , raw: stateless builders that fill the out response inside <code>on_railcom_request</code>

Migration note: the pre-refactor modules `dcc_application_main_track` and `dcc_application_service_track` still exist and are still compiled/tested alongside the role-first modules above; `dcc_config.c` does not wire them. Retiring them is tracked in *ComplianceOverview.md*. The third one, `dcc_packet_encoder`, was removed on 2026-09-23: it was an uncalled duplicate of

`dcc_application_command_station_packet` and had drifted (it still carried the `repeat_count = 0` "never transmitted" defect fixed in the live module).

6. Internal modules

Module	Role	Responsibility
<code>dcc_config</code>	always	Wiring: builds the interface structs of the modules it wires and owns their contexts (the service-mode task modules and the decoder-side modules keep their own static state), <code>initialize()/run()/ISR</code> dispatch
<code>dcc_types</code>	always	Typedefs, user-constant validation
<code>dcc_definitions</code>	always	Protocol constants: timing, instruction masks, CV numbers, RailCom IDs
<code>dcc_scheduler</code>	CS	Priority queue, duplicate combining, paced auto-refresh: a changed slot is sent <code>DCC_REFRESH_PROMPT_SENDS</code> times at full rate, then kept alive every <code>DCC_REFRESH_COLD_CYCLES</code> packet cycles and never later than <code>DCC_REFRESH_COLD_MAX_CYCLES</code> (selection order overdue, in-burst, due; <code>DCC_REFRESH_COLD_CYCLES = 0</code> is the flat round-robin ring); one-shots are sent <code>repeat_count</code> times (the builders set the <code>DCC_REPEAT_*</code> defaults; a one-shot with 0 is refused at insert). Pending one-shots always go before refresh slots and priority ranks only one-shots; the prompt burst is re-armed on every insert; an idle spacer separates back-to-back packets to the same short address 112–127
<code>dcc_bit_encoder</code>	CS	ISR bit framing from the shared fixed-period timer
<code>dcc_railcom_cutout</code>	CS + RAILCOM	RailCom cutout timer state machine
<code>dcc_railcom_command_station</code>	CS + RAILCOM	Receive drain after each cutout, Ch1/Ch2 datagram assembly from each byte's channel tag, a <code>dcc_railcom_result_enum</code> per channel, receive ring, tagging with loco addresses
<code>dcc_railcom_utilities</code>	RAILCOM, any role	4/8 code words (S-9.3.2 Table 2): encode for the decoder roles, decode for the command station
<code>dcc_service_mode_common</code>	CS	Shared ACK detection, reset sequencing, retry
<code>dcc_service_mode_{direct, paged, register, address}</code>	CS	Per-mode programming primitives
<code>dcc_service_mode_task_{direct, paged, register, address, detect}</code>	CS + own <code>TASK_*</code> flag	Read/write/verify orchestration on the primitives; mode detection
<code>dcc_bit_decoder</code>	DECODER	Edge-timestamp → bit classification → byte assembly

Module	Role	Responsibility
dcc_packet_decoder	DECODER	Parse bytes → structured commands, XOR, address match (own, broadcast, the CV19 consist address for speed/direction/e-stop and the functions CV21/CV22 enable; accessory board or output address for accessory packets), consist set/clear writes CV19, deferred dispatch queue
dcc_cv_storage	DECODER	CV abstraction, decoder lock, factory reset, indexed CVs, CV29 feature mask
dcc_fail_safe	DECODER	S-9.2.4 packet time-out (CV11 in 100 ms units)
dcc_railcom_decoder	DECODER + RAILCOM	RailCom transmit engine (bit-bang) and reply arming

Most modules own an `interface_dcc_<module>_t` of function pointers, populated by `dcc_config.c` (`dcc_railcom_utilities` and the packet builders have none, and the two legacy modules noted in §5 are not populated). This makes every dependency mockable in unit tests and lets an MCU swap touch only the config struct.

7. Execution contexts

Three contexts, with a deliberately small locked region:

- **ISR (timer / edge):** bit encoder advances the bit state machine and toggles the output pin; the 58 μ s tick also samples service-track current for ACK detection; RailCom cutout timing fires here. The decoder edge path classifies bits, assembles bytes, queues the finished packet, calls `on_railcom_request`, and bit-bangs the RailCom reply at 4 μ s intervals with interrupts masked (about 454 μ s; the CPU is otherwise idle then). ISR work is otherwise kept minimal.
- **Main loop (`DccConfig_run()`):** scheduler selection, packet build, service-mode state machine, RailCom receive drain, decoded-packet dispatch, fail-safe timing, 6 ms ACK-pulse timing. **Every command callback fires from this context; the decoder's `on_railcom_request` is the one exception.**
- **User API (any thread):** the `DccApplication*` calls write into shared state; on threaded hosts they must be protected by the user's `lock/unlock`.

The handoff between the ISR-level bit encoder and the main-loop scheduler is single-buffered: the encoder holds one active packet and a `packet_loaded` flag; the ISR clears the flag at the end bit and the main loop loads the next packet without taking a lock: a single producer and consumer hand off through the volatile flag, with `DCC_COMPILER_BARRIER()` keeping the packet stores ahead of the flag store on both sides (the only lock calls in the library surround the decoder's RailCom transmit). The cutout arm is deferred one tick so it lands on the end bit's last edge, because the encoder's state machine runs a half-bit ahead of the wire.

8. Key invariants

1. No direct cross-module includes in the core library — only via interface structs; the utilities modules are exempt (style guide).
2. Every new function/type is wrapped in the appropriate `DCC_COMPILE_*` guard.
3. No dynamic memory; all sizing comes from `USER_DEFINED_DCC_*` constants.
4. CV numbers are 1-based everywhere; the 0-based wire encoding is confined to the packet builders (`dcc_application_command_station_packet.c`), the service-mode primitives (`dcc_service_mode_direct.c`, `dcc_service_mode_paged.c`), and the packet decoder's conversion back.
5. Callbacks are main-loop only, except `on_railcom_request`; ISRs do pin/flag work and defer notification.